

ShopEz

Full Stack MERN E-Commerce Application

Project Technical Documentation

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1. Introduction

Project Title: ShopEz

Developer: INAPAKUTHIKA BHARATH VARMA (Solo Project)

ShopEz represents a sophisticated, end-to-end e-commerce environment engineered to provide consumers with a fluid and modern retail journey. Built upon the high-performance MERN stack (comprising MongoDB, Express.js, React.js, and Node.js), the application effectively links users to a diverse product catalog through a highly responsive and user-centric interface.

2. Project Overview

Purpose: The primary objective of ShopEz is to establish a secure, scalable platform for digital commerce. It facilitates intuitive product discovery, streamlined cart management, and safe transaction processing. Furthermore, it incorporates a specialized administrative layer designed for precise inventory oversight and lifecycle management of customer orders.

Key Features:

- **User Module:** Includes secure account registration, JWT-based login mechanisms, and personalized user profiles.
- **Product Module:** Features structured browsing across categories (Men, Women, Kids), alongside advanced search and filtering capabilities.
- **Cart & Checkout:** Enables dynamic, real-time shopping cart updates and supports Cash on Delivery (COD) as a primary payment method.
- **Admin Dashboard:** A comprehensive control center for managing product inventory, analyzing sales, and updating order fulfillment statuses.

3. System Architecture

Frontend (React.js):

The client-side is constructed using a Component-Based Architecture to ensure modularity and code reusability. Application state is managed efficiently via the React Context API or local state hooks. Navigation across the application is handled by React Router DOM, facilitating a Single Page Application (SPA) experience with client-side routing.

Backend (Node.js & Express.js):

The server-side follows an MVC-inspired design to provide a robust RESTful API. Express.js manages the routing logic, while custom middleware functions handle security, user authentication, and data validation. The backend serves structured JSON responses to interact seamlessly with the React frontend.

Database (MongoDB):

A NoSQL document-oriented database is utilized for flexible and high-speed data storage. The primary data collections include:

- **Users:** Manages secure credentials, contact details, and hierarchical roles (User vs. Admin).
- **Products:** Stores comprehensive metadata such as titles, descriptions, categories, pricing, and stock levels.
- **Orders:** Tracks complete transaction histories, item references, and logistical shipping statuses.

4. Setup Instructions

Prerequisites:

- Node.js (Version 18.x or newer)
- Package Managers: NPM or Yarn
- MongoDB Community Server (Configured to run on local port 27017)

Installation Steps:

1. Repository Initialization:

```
https://github.com/bharathvarma906/SHOPEZ_PROJECT.git
```

2. Server Configuration:

```
cd server && npm install
```

3. Client Configuration:

```
cd client && npm install
```

4. Environment Setup: Create a .env file within the server root directory with the following variables:

```
PORT=5000  
MONGO_URI=mongodb://localhost:27017/shopez  
JWT_SECRET=your_secure_secret_key
```

5. Project Organization

Directory	Description
/server/models	Data definitions using Mongoose schemas for Users, Products, and Orders.
/server/routes	Definitions for API endpoints and their corresponding logic handlers.
/client/src/components	Reusable UI elements like Header, Footer, and Product Cards.
/client/src/pages	High-level view components (Home, Product Details, Checkout).

6. Launching the System

To initialize the application in a local development environment, execute these commands in separate terminal instances:

Backend Service:

```
cd server && npm run dev
```

Frontend Application:

```
cd client && npm start
```

← → ↻ 🌐 localhost:3000

ShopEZ 🔍 Search products, brands, categories... 🏠 Home 📦 Products 🛒 Login

ShopEZ

Create Your Account

Join ShopEZ for seamless browsing, custom checkouts, and tracking panels.

FULL USER NAME
👤 Bharath Varma

EMAIL ADDRESS
✉️ bharath@gmail.com

PASSWORD (6+ CHARS)
🔒

CONFIRM PASSWORD
🔒

👤 REGISTER NEW ACCOUNT

Already have an account? [Sign In](#)

← → ↻ 🌐 localhost:3000

ShopEZ 🔍 Search products, brands, categories... 🏠 Home 📦 Products 🛒 3 Hello, Bharath Varma ➡ Logout



PRODUCT DESCRIPTION

Rugged dust-proof fitness companion containing high-precision blood oxygen sensors, continuous optical heart monitors, dual-band satellite GPS, and customized notifications.

Availability Status **In Stock (8 items left!)**

CHOOSE QUANTITY
- 1 + Max capacity: 8

[🛒 Add To Cart](#) [Buy Now](#)

🛡️ Guaranteed genuine product from authorized brand suppliers & merchants.

Share Your Review

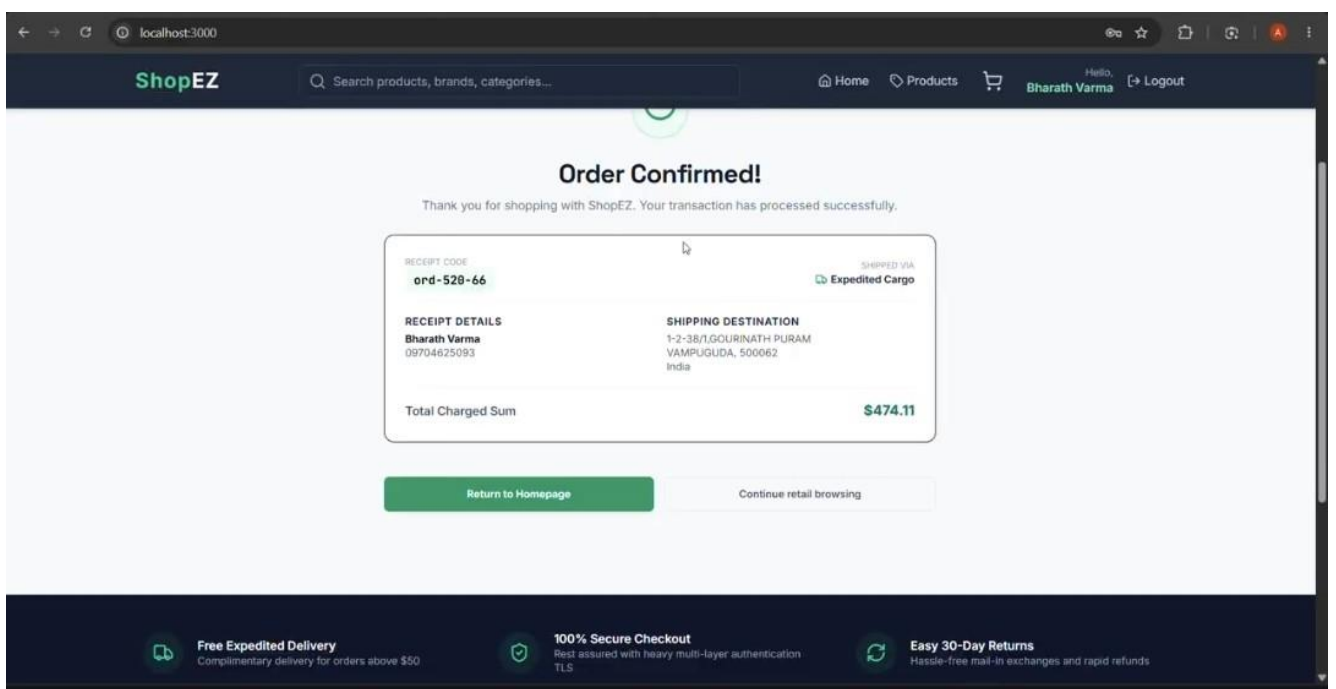
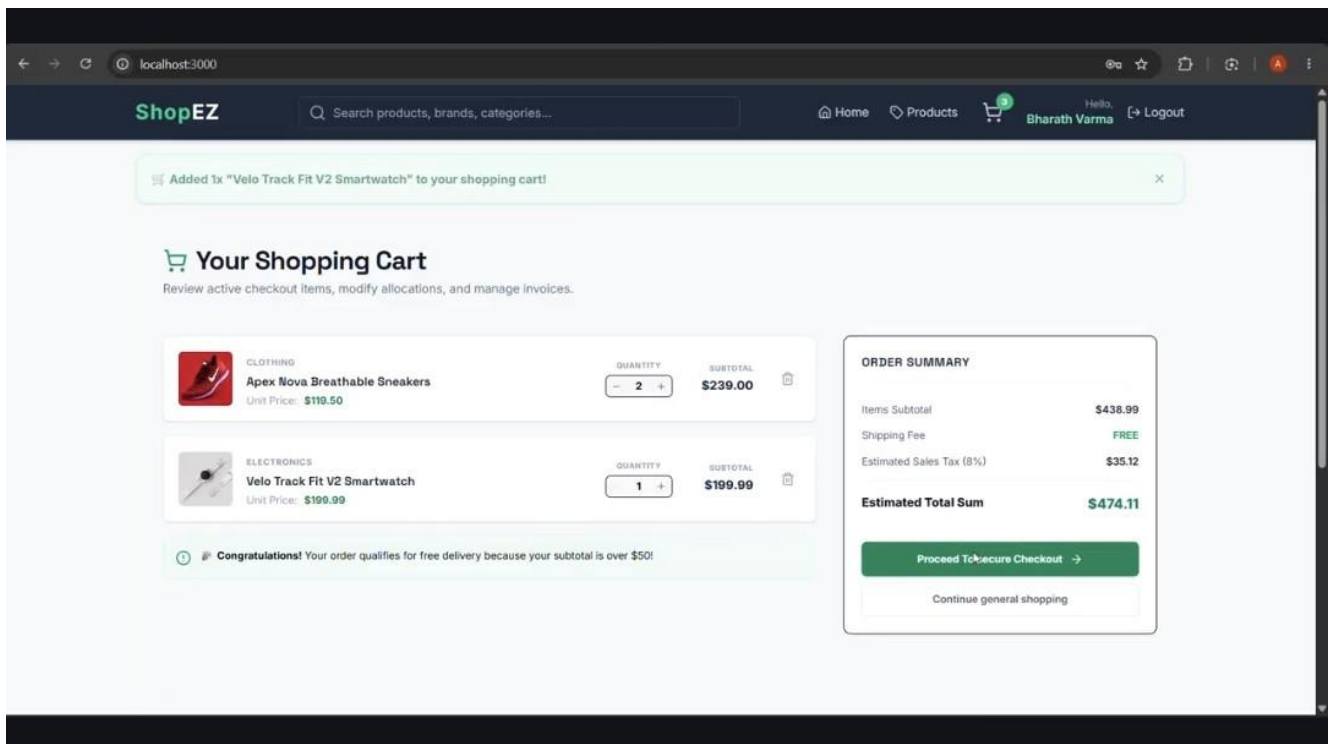
YOUR RATING
★★★★★

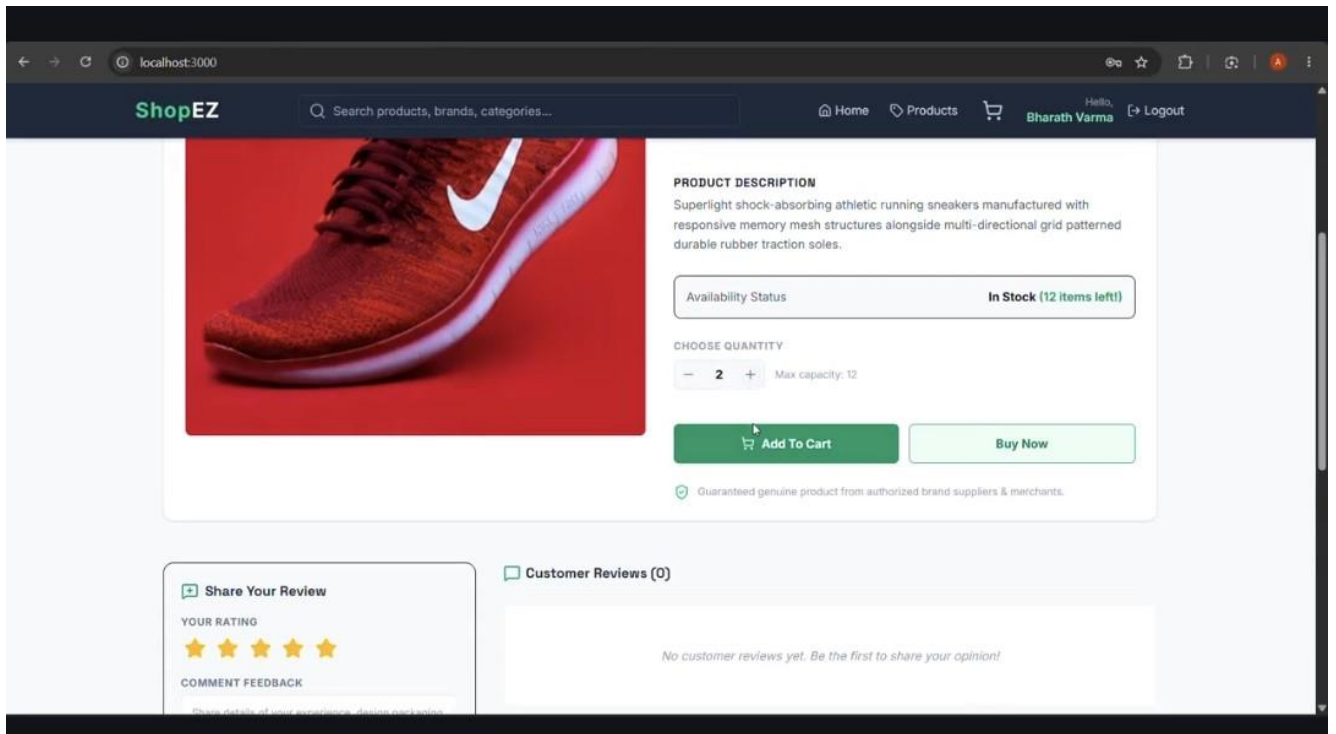
COMMENT FEEDBACK

Customer Reviews (1)

David Lee
★★★★★
🕒 5/29/2028

"Extremely accurate trails GPS tracker. Highly recommend the heart monitoring tools."





7. API Specifications

HTTP Method	Endpoint	Functionality
POST	/api/users/login	Validates user credentials and issues a JWT token.
GET	/api/products	Retrieves the product list with support for filtering criteria.
POST	/api/orders	Registers a new COD transaction (Authorized Access).

8. Security: Auth & Permissions

The ShopEz application employs JSON Web Tokens (JWT) for secure, stateless user sessions. Upon successful login, the server issues a signed token which the browser stores in LocalStorage or Cookies. This token is subsequently included in the 'Authorization' header for all protected requests. Custom middleware intercepts these requests to verify the token and enforce role-based access control, such as restricting access to the administrative dashboard.

9. UI Design Philosophy

The user interface adheres to a clean, minimalist design standard to prioritize ease of use. Key UI components include a sticky responsive navbar, grid-based product displays, and a streamlined checkout workflow. The design incorporates visual placeholders for high-resolution images and banners to ensure a professional aesthetic.

10. Quality Assurance & Testing

A structured testing plan was implemented to maintain system integrity:

- **Unit Testing:** focused on verifying the logic of standalone helper functions and data validators.
- **Integration Testing:** validated that Express API routes successfully communicate with the MongoDB database.
- **User Scenarios:** Examples include ensuring the system correctly blocks adding out-of-stock items to the cart and displays appropriate warnings.

11. Showcase & Repositories

Source Code & Demo Link: <https://github.com/bharathvarma906/SHOPEZ.git>

[Note: Integrated screenshots of the Landing Page, Administrator Panel, and Checkout Success screens to be placed here.]

12. Known Limitations

- Resetting client-side filters may occasionally require a manual browser refresh.
- Handling very large image files directly in MongoDB may affect load times; transitioning to a dedicated service like Cloudinary is planned.

13. Roadmap & Future Scope

- Integration of digital payment gateways such as Stripe or Razorpay.
- Development of an AI-driven system for personalized product suggestions.
- Enhancing the app into a Progressive Web App (PWA) for improved offline capabilities.